

ML Week 4 — Notation Cheat Sheet

Classification: Linear Classifiers, Logistic Regression & kNN

The ML Classification Pipeline



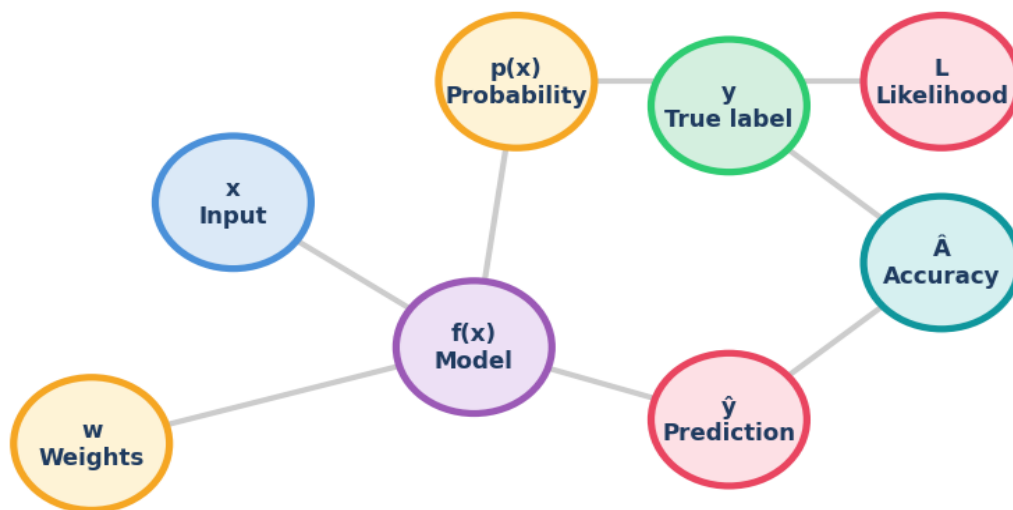
Every equation in the lecture serves one of these steps.

The Core Symbols

Symbol	Name	Plain English	Think of it as...
x	Input / predictors	The features we measure	"The stuff we know about each sample"
x_i	Sample i	The i -th data point	"Row i in the spreadsheet"
y	True label	The correct class	"The right answer"
$\hat{y}$ (y -hat)	Prediction	What the model guesses	"Our guess" — hat (^) = estimated
$f(x)$	Model / classifier	Function that predicts	"The prediction machine"
w	Weights	Model parameters	"How important each feature is"
$w \cdot x$	Dot product	Weights $\times$ features, summed	"Score: which side of the line?"
N	Sample count	Number of data points	"How many rows"
K	Neighbours (kNN)	Nearby points to check	"Ask K nearest friends to vote"
$\hat{A}$	Accuracy	$\text{\#correct} / \text{\#total}$	"What % did we get right?"
$\hat{E} = 1 - \hat{A}$	Error rate	$\text{\#wrong} / \text{\#total}$	"What % did we get wrong?"
$p(x)$	Probability	Certainty of classification	"How confident are we?"
$L = \prod p(x_i)$	Likelihood	Product of all certainties	"Multiply all confidences together"
$l = \sum \log p(x_i)$	Log-likelihood	Sum of log certainties	"Add logs instead of multiplying"

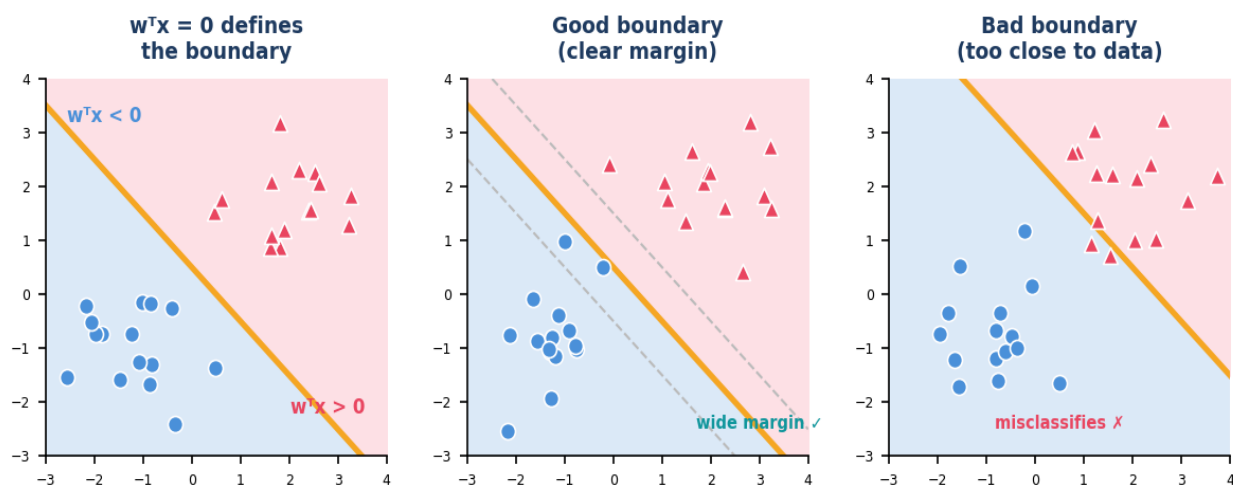
How The Symbols Connect

How All The Symbols Connect



Follow the connections — features (x) go into the model (f), which outputs predictions ($\hat{y}$), compared against truth (y) for accuracy ($\hat{A}$).

Linear Classifiers: Which Side of the Line?



The gold line is $w^T x = 0$. Left: basic split. Centre: well-placed with margin. Right: badly placed, misclassifies points.

■ Linear Classifier — Key Rules

- $w^T x = 0 \rightarrow$ you're ON the boundary
- $w^T x > 0 \rightarrow$ one side $\rightarrow$ predict class ■
- $w^T x < 0 \rightarrow$ other side $\rightarrow$ predict class ■
- Further from boundary = more confident
- **Linearly separable** = perfect line exists ($\hat{A} = 1$). **Non-separable** = best line still has errors.

Accuracy & Error Rate

$$\hat{A} = 8/10 = 80\% \quad \hat{E} = 2/10 = 20\% \quad \hat{E} = 1 - \hat{A}$$

✓ = correct ($\hat{y} = y$)

✗ = wrong ($\hat{y} \neq y$)



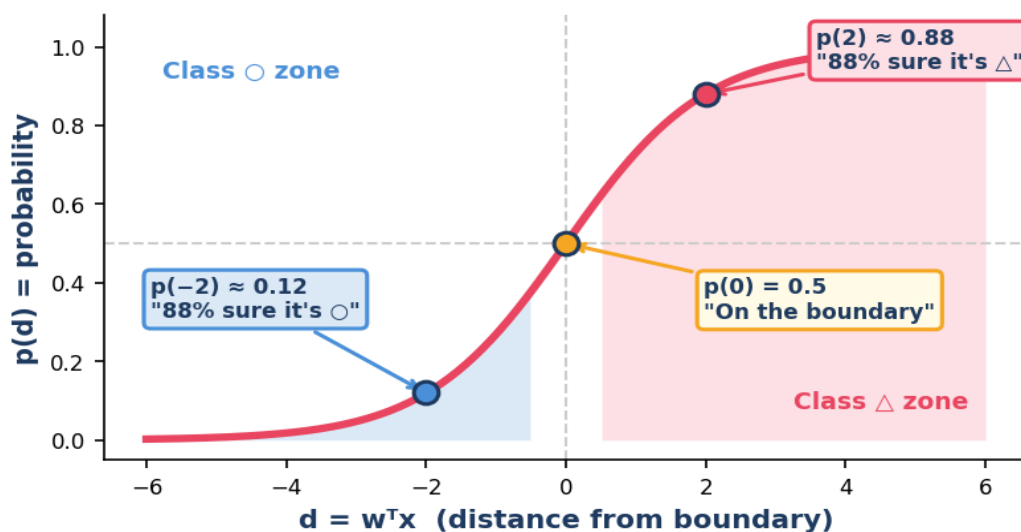
Count correct vs wrong. Accuracy = correct/total. Error = wrong/total = 1 – accuracy.

■ Symbol Shortcuts

- $\prod$ (capital Pi) = "multiply all" — just like Σ = "add all"
- $\hat{}$ (hat) = "estimated/predicted" — $\hat{y}$ is predicted y , $\hat{A}$ is estimated accuracy
- ^T (superscript T) = transpose — in practice just means "dot product"
- **log** = turns multiplication into addition (easier maths, same result)

The Logistic Function

The Logistic Function: Any Number → Probability (0 to 1)



The S-curve: any number in → probability out. Positive = likely ■, negative = likely ○, zero = 50/50.

■ Logistic Regression — What to Remember

- $p(d) = \frac{e^d}{1 + e^d}$ where $d = w^T x$
- $d = 0 \rightarrow p = 0.5$ (on the boundary, 50/50)
- d large positive $\rightarrow p \approx 1$ (confident it's ■)
- d large negative $\rightarrow p \approx 0$ (confident it's ○)
- Called "regression" because we regress on probability, not class directly
- Best logistic classifier **maximises likelihood L** , found via **gradient descent**

Likelihood: Quality Metric for Logistic Regression

Log-likelihood l = add $\log(\text{certainty})$ for each sample (same idea, easier maths)

Likelihood L = multiply all certainties together

$$L = 0.88 \times 0.92 \times 0.73 \times 0.95 = 0.561$$

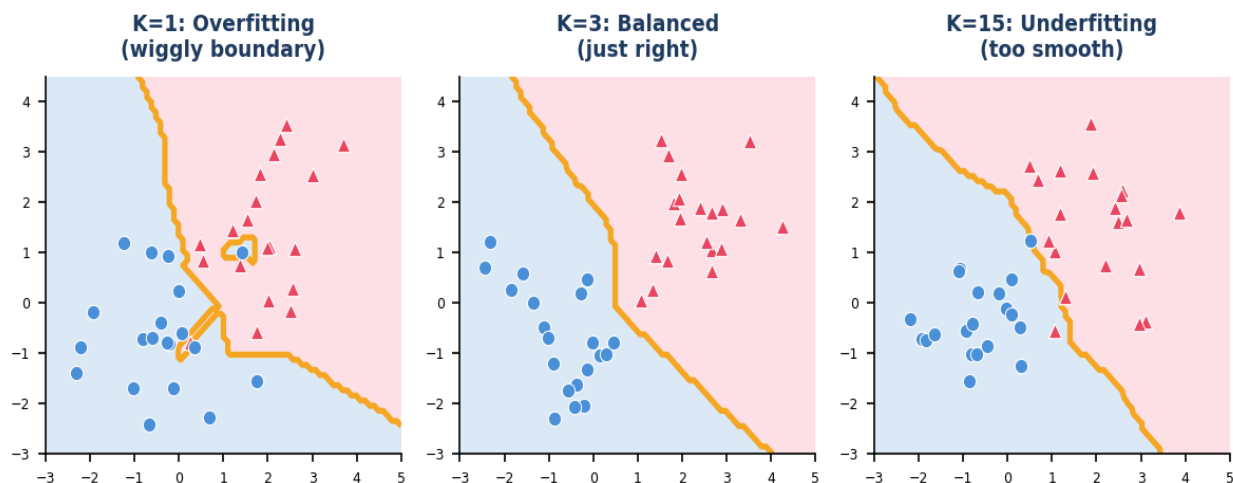


Each sample has a certainty. Multiply them all $\rightarrow$ higher L = better boundary.

Why Not Just Use Accuracy?

- Accuracy is binary — right or wrong, no "almost right"
- Likelihood measures HOW CONFIDENT the classifier is — richer metric
- Two classifiers can have same accuracy but different likelihoods
- Higher L = more confident overall = better model
- Logistic regression maximises L , not accuracy directly

k Nearest Neighbours (kNN)



$K=1$: overfits. $K=3$: balanced. $K=15$: underfits. Gold line = decision boundary.

kNN — Key Rules

- Classify new point: find K closest training samples, majority vote wins
- **Small K** $\rightarrow$ complex boundary $\rightarrow$ overfitting
- **Large K** $\rightarrow$ smooth boundary $\rightarrow$ underfitting
- Binary problems: use **odd K** to avoid ties
- **Non-parametric** — no assumed shape. **No training** — stores entire dataset.
- Called **instance-based** or **lazy** method

One-Page Quick Reference

■ NOTATION DECODER

- $\mathbf{x}_i$ = sample i | $\mathbf{y}_i$ = true label | $\hat{\mathbf{y}}_i$ = predicted label | N = total samples
- $\mathbf{w}$ = weights | $\mathbf{w} \cdot \mathbf{x}$ = dot product (score) | K = number of neighbours
- $\hat{A}$ = accuracy (#correct/#total) | $\hat{E}$ = error rate ($1 - \hat{A}$)
- $p(\mathbf{x})$ = logistic probability | L = likelihood ($\prod$ certainties) | I = log-likelihood ($\sum$ logs)
- Σ = add all | $\prod$ = multiply all | $\hat{}$ = estimated | T = transpose

■ THREE MODELS COMPARED

- **Linear classifier:** Straight-line boundary. $\mathbf{w} \cdot \mathbf{x} = 0$. Parametric. Simple but rigid.
- **Logistic regression:** Linear + probability. Maximises likelihood L via gradient descent.
- **kNN:** No assumed shape. Vote of K nearest. Non-parametric. No training. K controls complexity.

■ THE FLEXIBILITY TRADE-OFF

- **Too simple** (underfitting) → misses pattern → low accuracy on training AND test
- **Too complex** (overfitting) → memorises noise → high training, LOW test accuracy
- **Just right** → captures pattern, ignores noise → good on BOTH
- Linear: always same complexity. kNN: small K = complex, large K = simple.

■ EVERY EQUATION DOES ONE OF THREE THINGS

1. **CLASSIFY:** Which side of the line? → $\mathbf{w} \cdot \mathbf{x}$ (positive or negative)
2. **MEASURE QUALITY:** How many right? → $\hat{A}$, $\hat{E}$. How confident? → L , I
3. **FIND THE BEST MODEL:** Maximise L (logistic) or choose K (kNN)

The notation is just shorthand for simple ideas. You've got this. ■■

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